

ABSTRACT

The Library Management System is a web-based application developed to automate and simplify library operations. The system helps in managing books, members, transactions, returns, and fine calculations efficiently. The main objective of the project is to reduce manual work and improve accuracy in maintaining library records.

The project is developed using HTML, CSS, Bootstrap, JavaScript, PHP, and MySQL. The frontend is designed using HTML, CSS, and Bootstrap to create a user-friendly interface, while PHP is used for backend processing. MySQL database is used to store all records securely.

The system provides various features such as adding, editing, and deleting books and members, issuing and returning books, maintaining transaction history, and calculating fines automatically for delayed returns. QR code generation and QR code scanning functionalities are also implemented to make the issue and return process faster and more efficient.

Validation techniques are used to avoid duplicate entries and invalid data. The project improves efficiency, reduces paperwork, and helps in maintaining accurate records in libraries. The Library Management System is suitable for schools, colleges, and small organizations for managing library activities effectively.

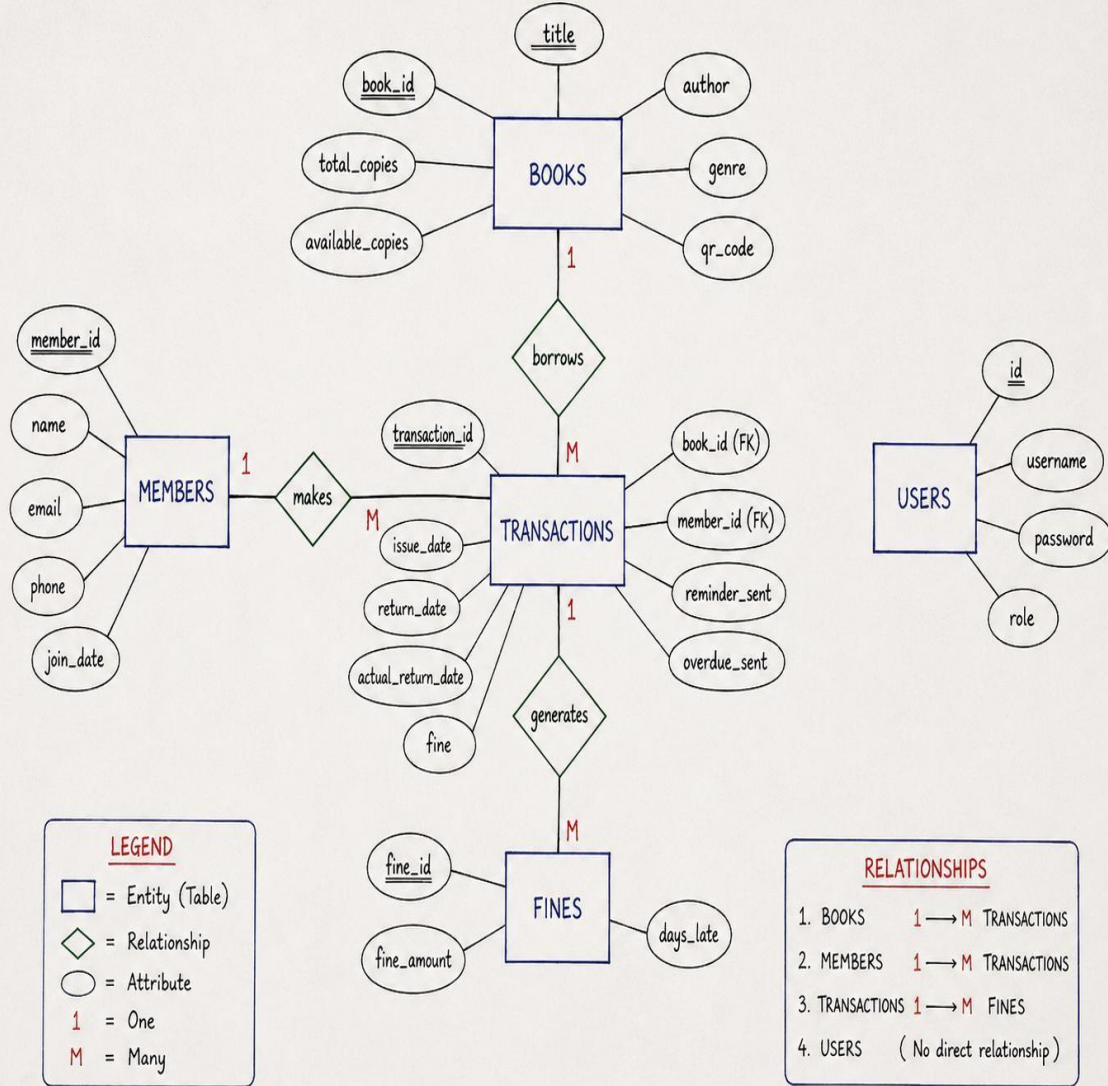
OBJECTIVES

The main objectives of the Library Management System are:

- To automate library operations and reduce dependency on manual record keeping.
- To maintain accurate and organized records of books available in the library.
- To manage member information efficiently and securely.
- To simplify the process of issuing and returning books.
- To track borrowing activities through transaction management.
- To calculate fines automatically for delayed book returns.
- To maintain separate records of fines and overdue transactions.
- To store and manage all library information using a MySQL database.
- To implement QR code generation and scanning for faster identification of books.
- To reduce data redundancy and improve consistency of information.
- To provide secure user authentication and access control. • To improve the speed and accuracy of library operations.
- To minimize paperwork and administrative effort.
- To provide quick search and retrieval of book and member records.
- To improve overall efficiency in managing library resources.
- To generate reliable records for monitoring library activities.
- To provide a user-friendly interface for librarians and administrators.
- To ensure secure storage and backup of library data.
- To support educational institutions and organizations in managing library services effectively.
- To enhance the quality and reliability of library management through automation.

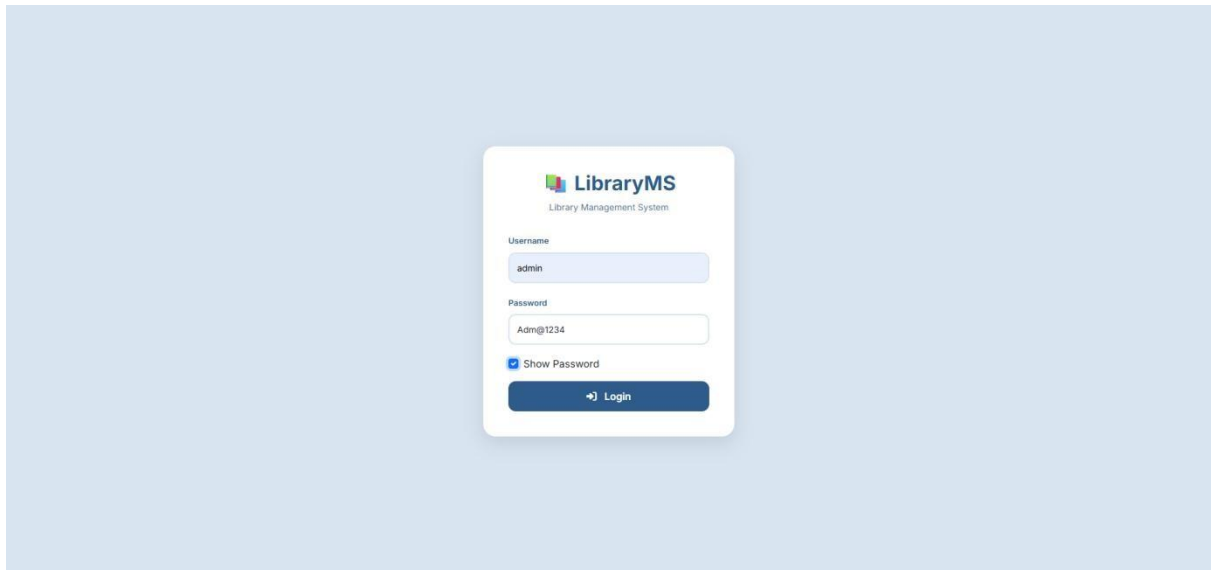
ER DIAGRAM

ER DIAGRAM - LIBRARY MANAGEMENT SYSTEM (5 TABLES)



OUTPUT SCREENS

LOGIN PAGE



The login page features a central white card on a light blue background. The card has the 'LibraryMS' logo at the top, followed by the text 'Library Management System'. Below this are input fields for 'Username' (containing 'admin') and 'Password' (containing 'Adm@1234'). A checkbox labeled 'Show Password' is checked. At the bottom of the card is a blue 'Login' button with a right-pointing arrow icon.

Figure 1: Login Page

This page is used for secure administrator login using username and password credentials.

USER PAGE



The user page has a light blue header with 'Welcome Student' and a 'Logout' button. Below is a table titled 'Available Books' with four columns: Title, Author, Genre, and Available Copies. The table lists 20 books with their respective details.

Title	Author	Genre	Available Copies
power of 100 mans	roholaan mie	action	21
rich dadd poor dad	Robert Kiyosaki	Finance	22
wings of fire	A P J Abdul Kalam	Biography	7
Atomic Habits	James Clear	Self Help	15
dbms concepts	korth	Education	27
war100	milten	action	35
7 habits	Rosh	Self Help	24
ad100	namdu	action	7
the life	mina	self help	5
Never Ever	Jhon Roy	Suspense	30
1 to 1	reena	philosophy	6
The Guide	R K Narayan	non fiction	12
1984	George Orwell	science fiction	10
The shining	Stephen King	Horror	10
To Kill a Mockingbird	Harper Lee	Fiction	11
The Hobbit	J tolkien	fantasy	15
It Ends with us	Coolten Hoover	Romance	15
2 states	Chetan Bhagat	Romance	10
The Immortals Of Meluha	Amish Tripathi	Mythological	11

Figure 2 : User Page

The Users table stores login credentials and role information of authorized users in the Library Management System.

DASHBOARD

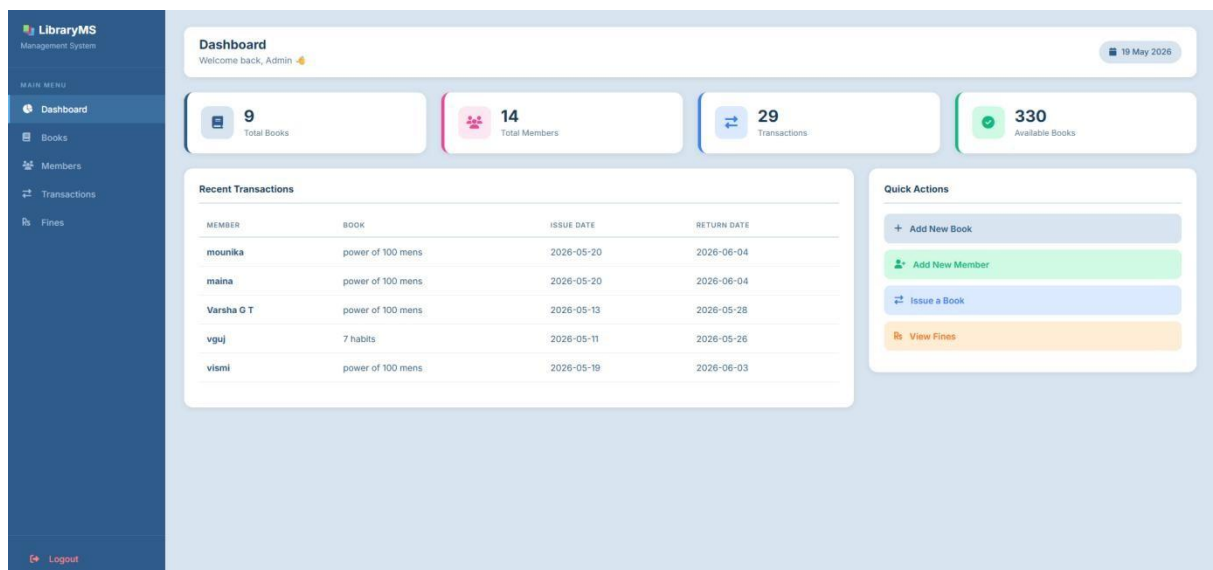


Figure 3 : Dashboard Page

This page displays the overall summary of the Library Management System including total books, members, transactions, and fines.

BOOKS PAGE

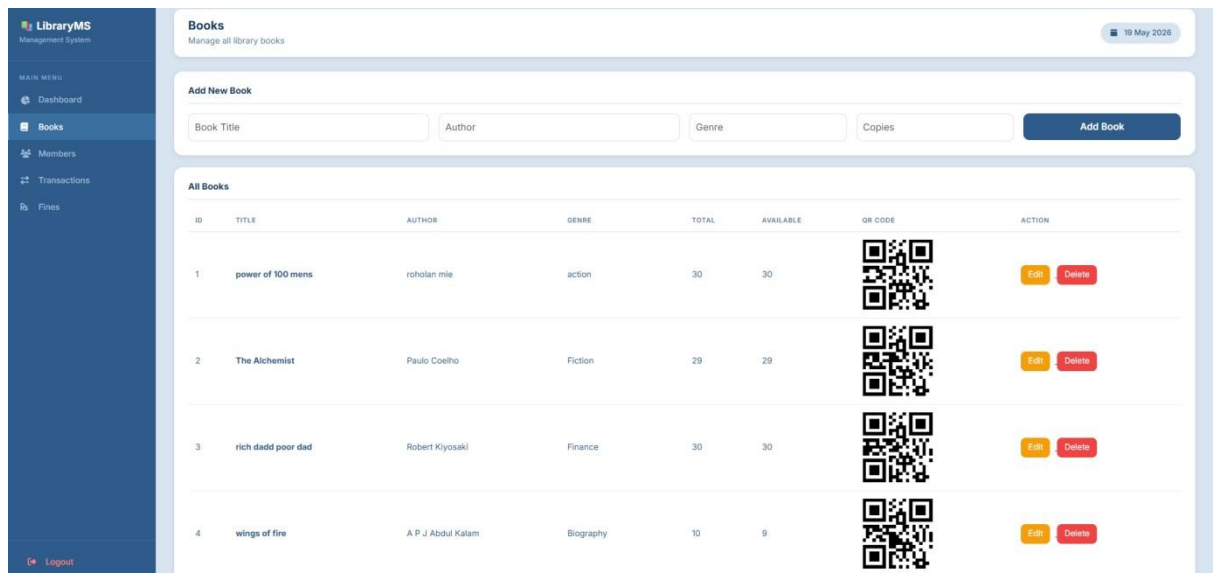
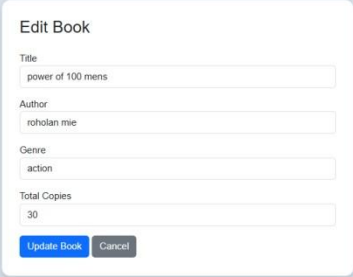


Figure 4: Books Page

This page is used for managing books in the library. The administrator can add, edit, delete, and view book details. QR codes are also generated for books

EDIT BOOK PAGE

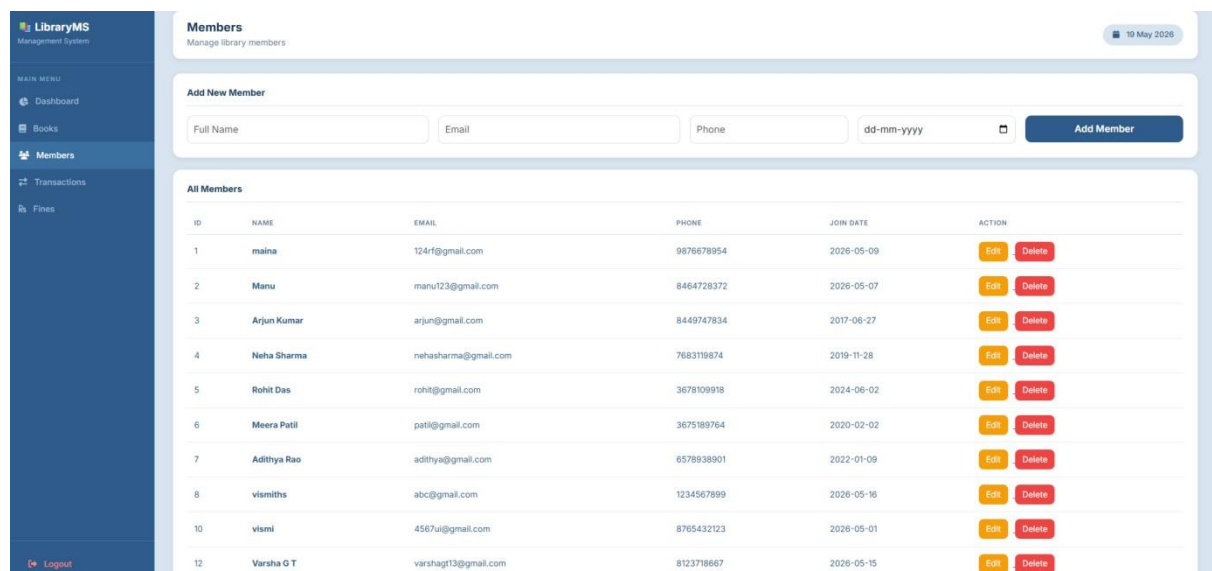


The image shows a modal form titled "Edit Book" with a light blue background. The form contains four input fields: "Title" with the value "power of 100 mens", "Author" with "roholaan mie", "Genre" with "action", and "Total Copies" with "30". At the bottom of the form are two buttons: "Update Book" in blue and "Cancel" in grey.

Figure 6: Edit Book Page

This page is used for updating existing book details. Validation techniques are implemented to avoid invalid book information.

MEMBERS PAGE

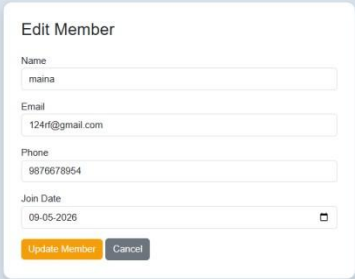


The image shows the "Members" page of the LibraryMS Management System. It features a sidebar with navigation links: Dashboard, Books, Members (active), Transactions, and Fines. The main content area has a header "Members" with the subtitle "Manage library members" and a date "10 May 2026". Below the header is a form to "Add New Member" with fields for "Full Name", "Email", "Phone", and "dd-mm-yyyy", followed by an "Add Member" button. The main section is titled "All Members" and contains a table with 12 rows of member data. Each row has columns for ID, NAME, EMAIL, PHONE, JOIN DATE, and ACTION (with Edit and Delete buttons).

ID	NAME	EMAIL	PHONE	JOIN DATE	ACTION
1	maina	124r1@gmail.com	9876678954	2026-05-09	Edit Delete
2	Manu	manu123@gmail.com	8464728372	2026-05-07	Edit Delete
3	Arjun Kumar	arjun@gmail.com	8449747834	2017-06-27	Edit Delete
4	Neha Sharma	nehasharma@gmail.com	768319874	2019-11-28	Edit Delete
5	Rohit Das	rohit@gmail.com	3678109918	2024-06-02	Edit Delete
6	Meera Patil	patil@gmail.com	367599764	2020-02-02	Edit Delete
7	Adithya Rao	adithya@gmail.com	6578938901	2022-01-09	Edit Delete
8	vismiths	abc@gmail.com	1234567899	2026-05-16	Edit Delete
10	vismi	4567ui@gmail.com	8765432123	2026-05-01	Edit Delete
12	Varsha G T	varshag13@gmail.com	8123718667	2026-05-15	Edit Delete

Figure 7: Members Page

This page is used for managing library members. The administrator can add, edit, delete, and view member details.



Edit Member

Name
maina

Email
124rf@gmail.com

Phone
9676678954

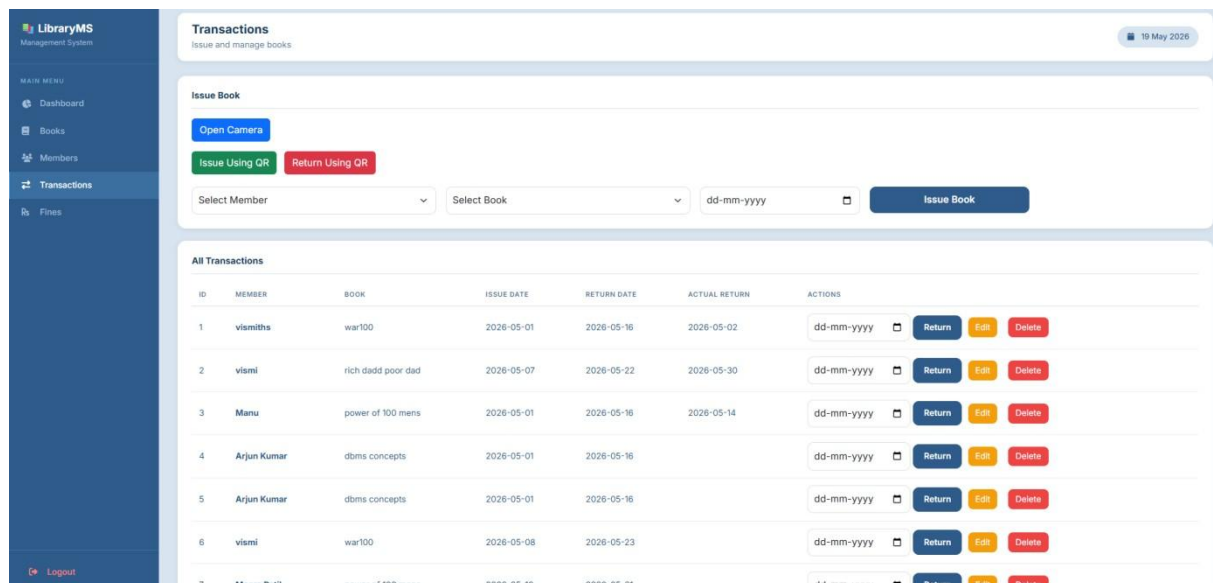
Join Date
09-05-2026

[Update Member](#) [Cancel](#)

Figure 8: Edit Member Page

This page is used for updating member details. Duplicate email and phone number validations are implemented in this module.

TRANSACTIONS PAGE



LibraryMS
Management System

MAIN MENU
Dashboard
Books
Members
Transactions
Fines
Logout

Transactions
Issue and manage books

19 May 2026

Issue Book

[Open Camera](#)
[Issue Using QR](#) [Return Using QR](#)

Select Member Select Book dd-mm-yyyy [Issue Book](#)

All Transactions

ID	MEMBER	BOOK	ISSUE DATE	RETURN DATE	ACTUAL RETURN	ACTIONS
1	vismiths	war100	2026-05-01	2026-05-16	2026-05-02	dd-mm-yyyy Return Edit Delete
2	vismi	rich dadd poor dad	2026-05-07	2026-05-22	2026-05-30	dd-mm-yyyy Return Edit Delete
3	Manu	power of 100 mens	2026-05-01	2026-05-16	2026-05-14	dd-mm-yyyy Return Edit Delete
4	Arjun Kumar	dbms concepts	2026-05-01	2026-05-16		dd-mm-yyyy Return Edit Delete
5	Arjun Kumar	dbms concepts	2026-05-01	2026-05-16		dd-mm-yyyy Return Edit Delete
6	vismi	war100	2026-05-08	2026-05-23		dd-mm-yyyy Return Edit Delete
7	Meera Patil	power of 100 mens	2026-05-16	2026-05-31		dd-mm-yyyy Return Edit Delete

Figure 9: Transactions Page

This page is used for managing book issue and return operations. The system stores transaction details such as issue date, return date, and actual return date.

QR CODE SCANNING

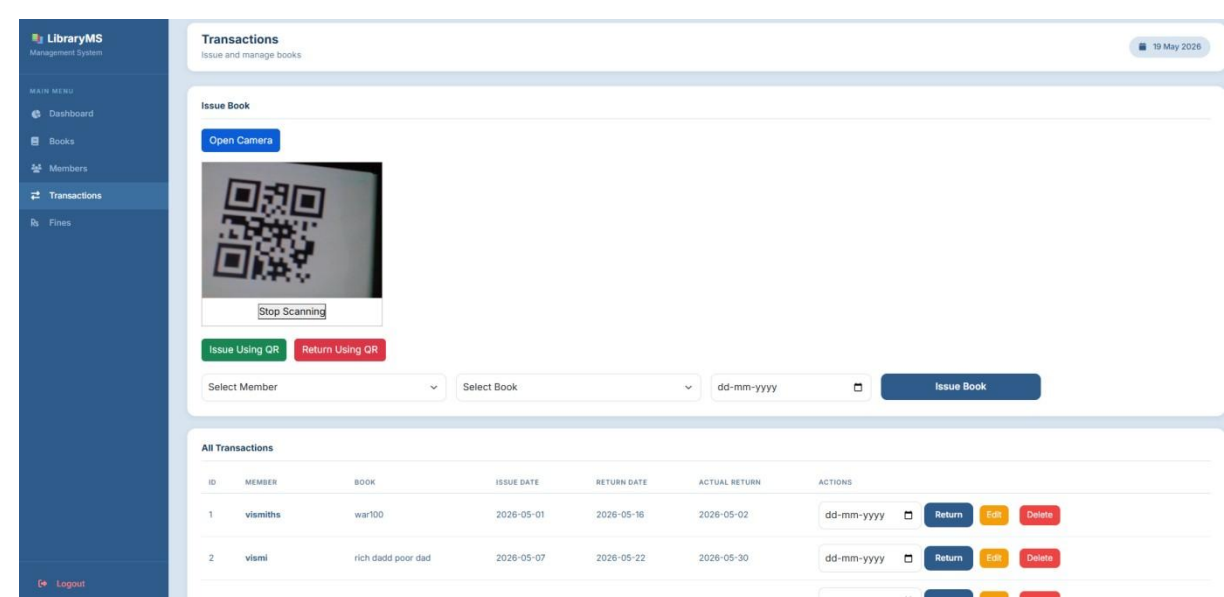


Figure 10: QR Code Scanning

This page demonstrates the QR code scanning feature used for automatically selecting books during transactions.

RETURN BOOK PAGE

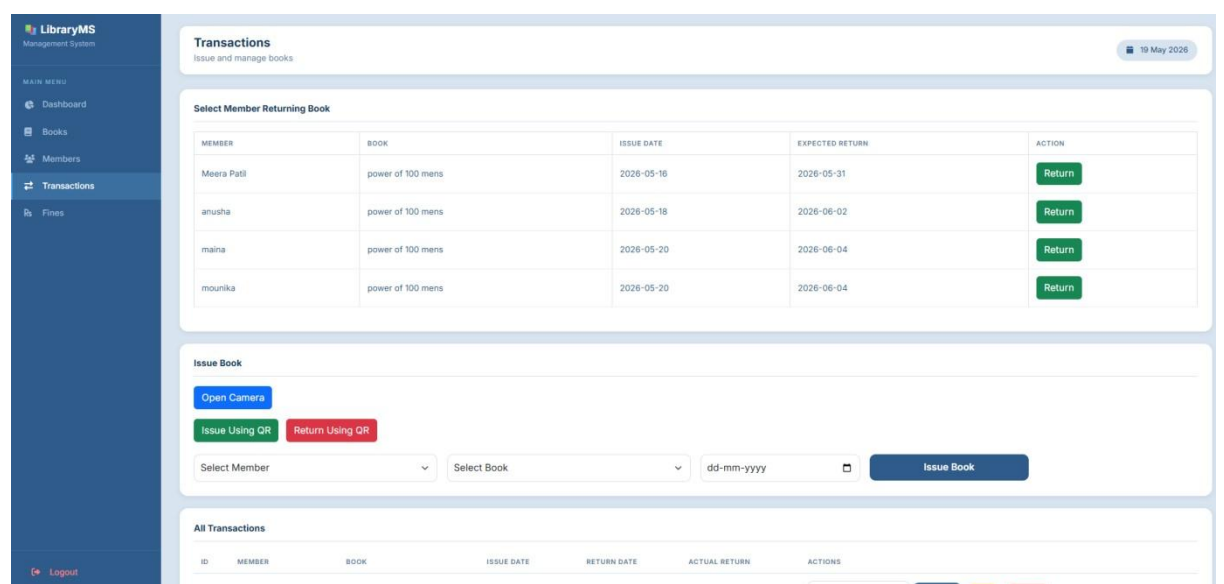
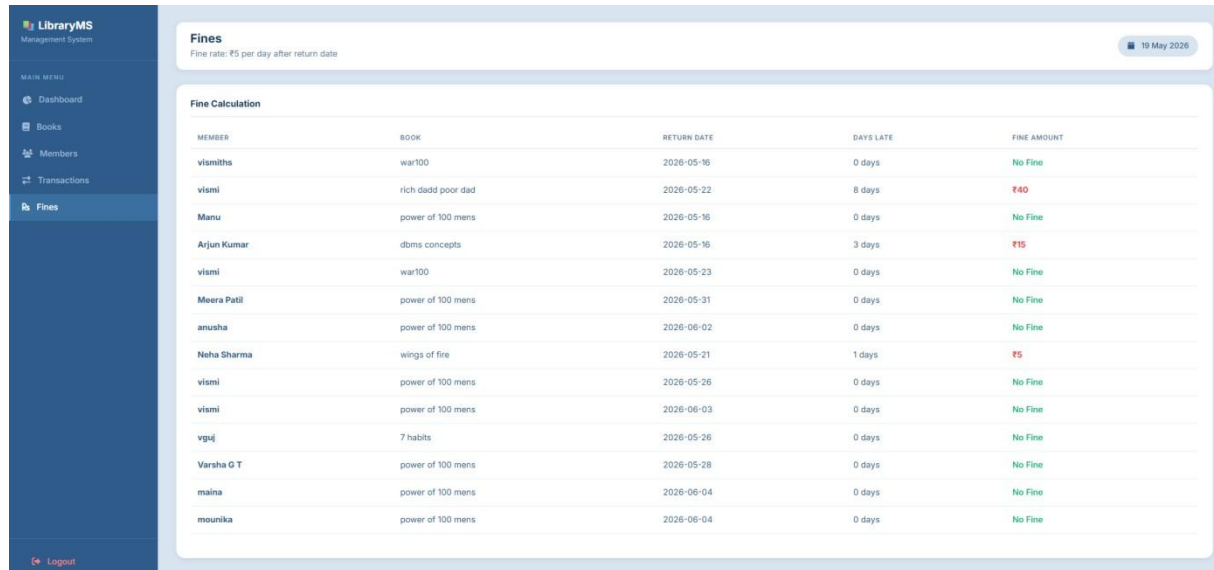


Figure 11: Return Book Page

This page is used for returning issued books. The system updates transaction records and calculates fines for delayed returns.

FINES PAGE



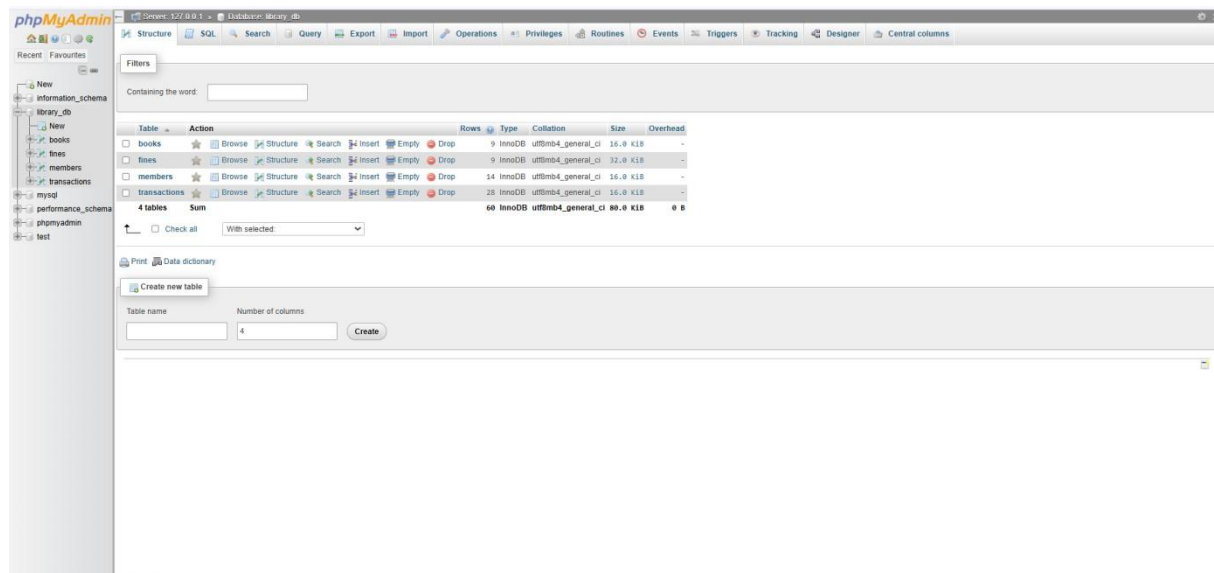
The screenshot shows the 'Fines' page in the LibraryMS system. The page header indicates a fine rate of ₹5 per day after the return date. The main content is a table titled 'Fine Calculation' with columns: MEMBER, BOOK, RETURN DATE, DAYS LATE, and FINE AMOUNT. The table lists 15 entries for various members and books, showing the return date, the number of days late, and the corresponding fine amount. Some entries show 'No Fine' while others show a specific amount in red text.

MEMBER	BOOK	RETURN DATE	DAYS LATE	FINE AMOUNT
vismiths	war100	2026-05-16	0 days	No Fine
vismi	rich dadd poor dad	2026-05-22	8 days	₹40
Manu	power of 100 mens	2026-05-16	0 days	No Fine
Arjun Kumar	dbms concepts	2026-05-16	3 days	₹15
vismi	war100	2026-05-23	0 days	No Fine
Meera Patil	power of 100 mens	2026-05-31	0 days	No Fine
anusha	power of 100 mens	2026-06-02	0 days	No Fine
Neha Sharma	wings of fire	2026-05-21	1 days	₹5
vismi	power of 100 mens	2026-05-26	0 days	No Fine
vismi	power of 100 mens	2026-06-03	0 days	No Fine
vguj	7 habits	2026-05-26	0 days	No Fine
Varsha G T	power of 100 mens	2026-05-28	0 days	No Fine
maina	power of 100 mens	2026-06-04	0 days	No Fine
mounika	power of 100 mens	2026-06-04	0 days	No Fine

Figure 12: Fines Page

This page displays fine details for overdue book returns including delayed days and fine amount.

DATABASE PAGE



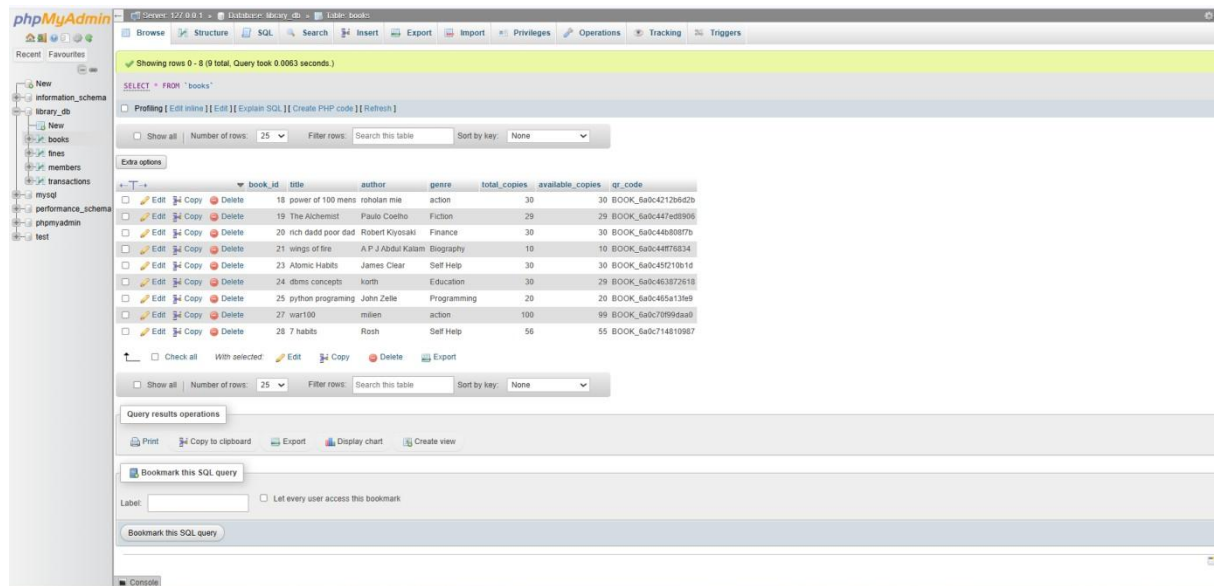
The screenshot shows the phpMyAdmin interface for the 'library_db' database. The left sidebar shows the database structure with tables: books, fines, members, transactions, performance_schema, and phpmyadmin. The main area displays the 'Structure' tab for the 'library_db' database, showing a list of tables with their row counts, types, collations, sizes, and overheads. Below the table list, there is a 'Create new table' section with fields for 'Table name' and 'Number of columns' (set to 4), and a 'Create' button.

Table	Rows	Type	Collation	Size	Overhead
books	9	InnoDB	utf8mb4_general_ci	16.0 K B	-
fines	9	InnoDB	utf8mb4_general_ci	32.0 K B	-
members	14	InnoDB	utf8mb4_general_ci	16.0 K B	-
transactions	28	InnoDB	utf8mb4_general_ci	16.0 K B	-
4 tables Sum	60	InnoDB	utf8mb4_general_ci	80.0 K B	0 B

Figure 13: Database Tables in phpMyAdmin

This page displays the MySQL database tables used in the Library Management System including books, members, transactions, and fines tables.

BOOKS TABLE DATABASE



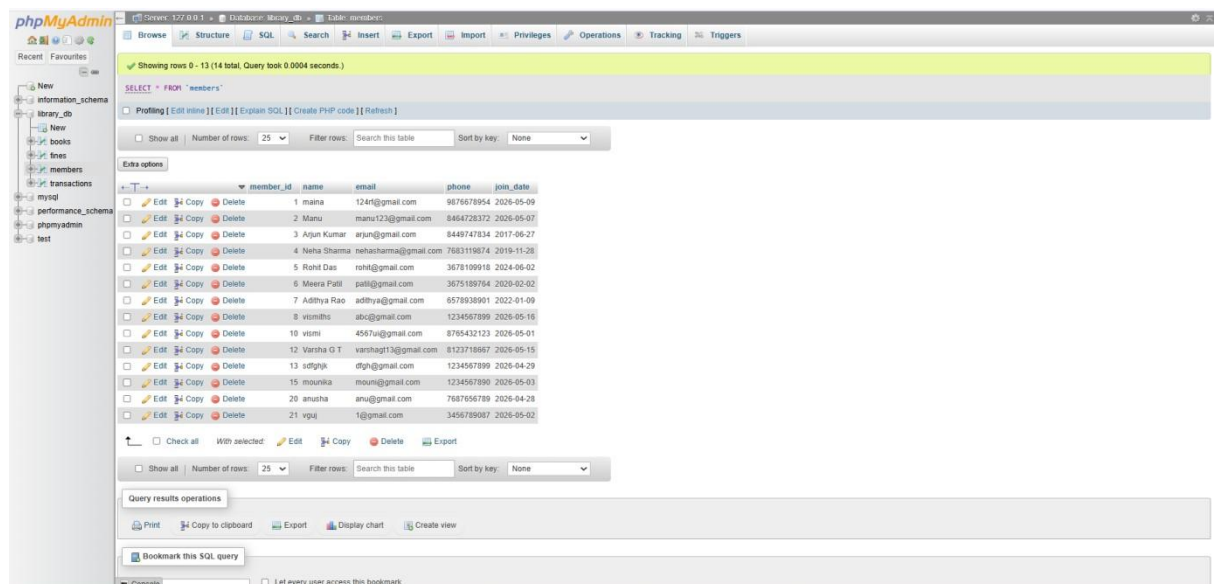
The screenshot shows the phpMyAdmin interface for the 'library_db' database. The 'books' table is selected, and its structure is displayed. The table has 8 columns: book_id, title, author, genre, total_copies, available_copies, qr_code, and a primary key. The data is shown in a table with 10 rows. Below the table, there are options to perform query results operations like Print, Copy to clipboard, Export, Display chart, and Create view. There is also a section to bookmark the SQL query.

book_id	title	author	genre	total_copies	available_copies	qr_code
18	power of 100 men	rohlan mie	action	30	30	BOOK_5a0c4212642b
19	The Alchemist	Paulo Coelho	Fiction	29	29	BOOK_5a0c447e0909
20	rich dad poor dad	Robert Kiyosaki	Finance	30	30	BOOK_5a0c44b8087b
21	wings of fire	A P J Abdul Kalam	Biography	10	10	BOOK_5a0c4487f034
23	Atomic Habits	James Clear	Self Help	30	30	BOOK_5a0c450210b16
24	400 concepts	korth	Education	30	29	BOOK_5a0c453872618
25	python programming	John Zelle	Programming	20	20	BOOK_5a0c465a139e8
27	war100	milen	action	100	99	BOOK_5a0c70994aa8
28	7 habits	Rosh	Self Help	56	55	BOOK_5a0c714810987

Figure 14: Books Table

This table stores book-related information such as title, author, genre, and available copies.

MEMBERS TABLE DATABASE



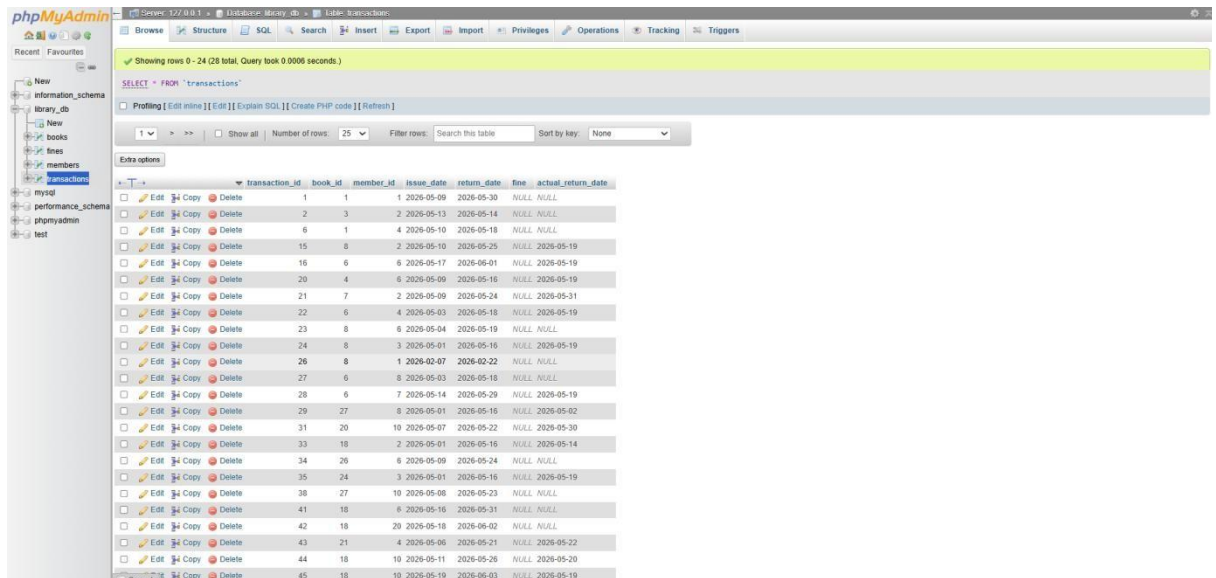
The screenshot shows the phpMyAdmin interface for the 'library_db' database. The 'members' table is selected, and its structure is displayed. The table has 5 columns: member_id, name, email, phone, and join_date. The data is shown in a table with 21 rows. Below the table, there are options to perform query results operations like Print, Copy to clipboard, Export, Display chart, and Create view. There is also a section to bookmark the SQL query.

member_id	name	email	phone	join_date
1	maina	124t@gmail.com	9876678954	2026-05-09
2	Manu	manu123@gmail.com	8464726372	2026-05-07
3	Arun Kumar	arjun@gmail.com	8449747834	2017-06-27
4	Neha Sharma	nehasharma@gmail.com	7683118674	2019-11-28
5	Rohit Das	rohit@gmail.com	3678109918	2024-06-02
6	Meera Patel	patil@gmail.com	3675189764	2020-02-02
7	Adithya Rao	adithya@gmail.com	6578938901	2022-01-09
8	vismiths	abc@gmail.com	1234567899	2026-05-16
10	vismi	4567ui@gmail.com	8785432123	2026-05-01
12	Varsha G T	varshagt13@gmail.com	8123718967	2026-05-15
13	sdghjk	dgh@gmail.com	1234567899	2026-04-29
15	mounika	moun@gmail.com	1234567899	2026-05-03
20	anusha	anu@gmail.com	7687666789	2026-04-28
21	vgsd	1@gmail.com	3456789087	2026-05-02

Figure 15: Members Table

This table stores member information such as name, email, phone number, and joining date.

TRANSACTIONS TABLE DATABASE

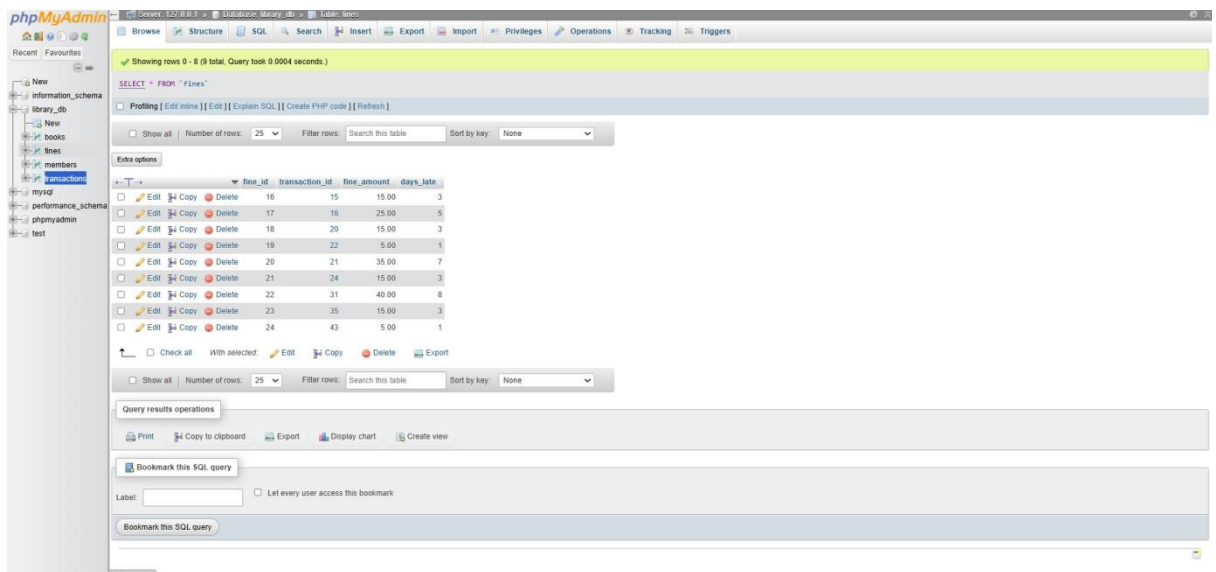


transaction_id	book_id	member_id	issue_date	return_date	fine	actual_return_date
1	1	1	2026-05-09	2026-05-30	NULL	NULL
2	3	2	2026-05-13	2026-05-14	NULL	NULL
6	1	4	2026-05-10	2026-05-19	NULL	NULL
15	8	2	2026-05-10	2026-05-25	NULL	2026-05-19
16	6	6	2026-05-17	2026-06-01	NULL	2026-05-19
20	4	6	2026-05-09	2026-05-16	NULL	2026-05-19
21	7	2	2026-05-09	2026-05-24	NULL	2026-05-31
22	6	4	2026-05-03	2026-05-16	NULL	2026-05-19
23	8	6	2026-05-04	2026-05-19	NULL	NULL
24	8	3	2026-05-01	2026-05-16	NULL	2026-05-19
26	8	1	2026-02-07	2026-02-22	NULL	NULL
27	6	8	2026-05-03	2026-05-16	NULL	NULL
28	6	7	2026-05-14	2026-05-29	NULL	2026-05-19
29	27	8	2026-05-01	2026-05-16	NULL	2026-05-02
31	20	10	2026-05-07	2026-05-22	NULL	2026-05-30
33	18	2	2026-05-01	2026-05-16	NULL	2026-05-14
34	26	6	2026-05-09	2026-05-24	NULL	NULL
35	24	3	2026-05-01	2026-05-16	NULL	2026-05-19
38	27	10	2026-05-08	2026-05-23	NULL	NULL
41	18	6	2026-05-16	2026-05-31	NULL	NULL
42	18	20	2026-05-18	2026-06-02	NULL	NULL
43	21	4	2026-05-06	2026-05-21	NULL	2026-05-22
44	18	10	2026-05-11	2026-05-26	NULL	2026-05-20
45	18	10	2026-05-19	2026-06-03	NULL	2026-05-19

Figure 16: Transactions Table

This table stores transaction details related to issued and returned books.

FINES TABLE DATABASE



fine_id	transaction_id	fine_amount	days_late
16	15	15.00	3
17	16	25.00	5
18	20	15.00	3
19	22	5.00	1
20	21	35.00	7
21	24	15.00	3
22	31	40.00	8
23	35	15.00	3
24	43	5.00	1

Figure 17: Fines Table

This table stores fine details including delayed days and fine amount for overdue book returns.

ALERT / VALIDATION MESSAGE SCREENSHOT

The screenshot shows the LibraryMS Transactions page. A green banner at the top states "Book Issued Successfully". Below this, the "Issue Book" section contains buttons for "Open Camera", "Issue Using QR", and "Return Using QR". There are also dropdown menus for "Select Member" and "Select Book", a date input field, and an "Issue Book" button. The "All Transactions" table lists five transactions with columns for ID, Member, Book, Issue Date, Return Date, Actual Return, and Actions.

ID	MEMBER	BOOK	ISSUE DATE	RETURN DATE	ACTUAL RETURN	ACTIONS
1	vismiths	war100	2026-05-01	2026-05-16	2026-05-02	dd-mm-yyyy [Return] [Edit] [Delete]
2	vismi	rich dadd poor dad	2026-05-07	2026-05-22	2026-05-30	dd-mm-yyyy [Return] [Edit] [Delete]
3	Manu	power of 100 mens	2026-05-01	2026-05-16	2026-05-14	dd-mm-yyyy [Return] [Edit] [Delete]
4	Arjun Kumar	dbms concepts	2026-05-01	2026-05-16	2026-05-19	dd-mm-yyyy [Return] [Edit] [Delete]
5	vismi	war100	2026-05-08	2026-05-23		dd-mm-yyyy [Return] [Edit] [Delete]

The screenshot shows the LibraryMS Transactions page with a red banner at the top stating "Member already has this book". The "Issue Book" section and "All Transactions" table are identical to the previous screenshot.

ID	MEMBER	BOOK	ISSUE DATE	RETURN DATE	ACTUAL RETURN	ACTIONS
1	vismiths	war100	2026-05-01	2026-05-16	2026-05-02	dd-mm-yyyy [Return] [Edit] [Delete]
2	vismi	rich dadd poor dad	2026-05-07	2026-05-22	2026-05-30	dd-mm-yyyy [Return] [Edit] [Delete]
3	Manu	power of 100 mens	2026-05-01	2026-05-16	2026-05-14	dd-mm-yyyy [Return] [Edit] [Delete]
4	Arjun Kumar	dbms concepts	2026-05-01	2026-05-16	2026-05-19	dd-mm-yyyy [Return] [Edit] [Delete]
5	vismi	war100	2026-05-08	2026-05-23		dd-mm-yyyy [Return] [Edit] [Delete]

Figure 18: Validation Alert Message

This page displays validation messages used to avoid duplicate entries and invalid transactions in the system.

CONCLUSION

The Library Management System successfully automates library operations such as managing books, members, transactions, and fines. The project reduces manual work and improves efficiency and accuracy in maintaining records.

The system is developed using HTML, CSS, Bootstrap, JavaScript, PHP, and MySQL. QR code generation and scanning features improve the speed and efficiency of transactions.

Validation techniques are implemented to avoid invalid and duplicate records. The project provides a user-friendly interface and secure database management.

The Library Management System is suitable for schools, colleges, and small organizations for managing library activities effectively and efficiently.